

Api Gateways

AWS API Gateway offers several key features:

- Support for multiple API types, including WebSocket (stateful) and HTTP/REST (stateless) APIs, along with flexible authentication options^s
- API versioning capabilities and performance monitoring through Amazon CloudWatch^s
- Traffic management functionality and integration with AWS security services to protect your APIs^s
- Cost-effective pricing structure where you only pay for actual API usage, with no upfront costs^s
- Low latency performance through Amazon's global network infrastructure^s

The service enables developers to create, publish, and maintain APIs that can access various AWS services or other web services, particularly RESTful APIs

Introduction to AWS and APIs

- Amazon Web Services (AWS) is the world's largest cloud service provider, offering over 175 services for different types of customers, including startups, enterprises, and government agencies^s.
- An API, or Application Programming Interface, is a software intermediary that enables two applications to communicate, making processes like online bookings efficient^s.

AWS API Gateway

- AWS API Gateway is a service for creating, publishing, maintaining, monitoring, and securing various APIs at any scale^s.
- It allows developers to create APIs accessing AWS services or other web services, including RESTful APIs which follow the principles of Representational State Transfer (REST)^s.

Features of AWS API Gateway

- Supports both WebSocket (stateful) and HTTP/REST (stateless) APIs, providing flexible authentication mechanisms^s.
- Offers functionality for API versioning, performance monitoring through Amazon CloudWatch, and traffic management^s.
- Provides cost-effective pricing and integration with AWS security services to protect APIs^{ss}.

Benefits of AWS API Gateway

- Facilitates efficient API development with features such as simultaneous versioning and a pricing model based on usage^{ss}.
- Ensures low latency by utilizing Amazon's global network and allows easy monitoring of performance metrics^s.

Types of APIs

- Discusses three kinds of APIs: REST APIs (stateless), WebSocket APIs (stateful), and HTTP APIs (lightweight and cost-effective)^s.

Key Terms and Concepts

- Definitions of key terms such as API development, API endpoints, API keys, stages, mapping templates, and methods to understand API functionalities^{ss}.

AWS API Gateway Pricing

- Users pay only for API calls and data transfers with no upfront fees. Free tier services are also provided for new users^{ss}.

Hands-On Demonstration: Creating a REST API Using AWS API Gateway

- Step-by-step instructions to create a Lambda function and connect it with AWS API Gateway to create a REST API^{ss}.
- Demonstration of creating another Lambda function and adding new methods, including mock integrations for independent API developmen

1. What is AWS?
2. What does API stand for and what is its purpose?
3. What is the primary function of the AWS API Gateway?
4. What does REST stand for in the context of APIs, and what is its significance?
5. What does REST stand for in the context of AWS API Gateway?
6. What are some key features of AWS API Gateway?
7. What pricing model does the Amazon API Gateway use for API requests?
8. How does Amazon API Gateway help manage traffic spikes?
9. What are the primary benefits of using an API Gateway?
10. Who are the two main types of developers that utilize an API Gateway?
11. What is a REST API in the context of AWS API Gateway?
12. What distinguishes a WebSocket API from a REST API in AWS API Gateway?
13. What is an API stage in AWS API Gateway?
14. What is a mapping template in AWS API Gateway?
15. What are the key components included in the AWS API Gateway free tier?
16. What factors determine the charges after the AWS API Gateway free tier ends?
17. What is the first step to create a REST API using AWS Lambda?
18. What does AWS Lambda do in terms of resource management?
19. What is the first step to create a Lambda function in AWS?
20. How do you define a successful response in a mock integration?What is the purpose of testing a Lambda function in AWS API Gateway?
21. How can you simulate a GET request to test your Lambda function in API Gateway?
22. What are the steps to install curl on a Windows operating system?
23. How do you create a REST API using AWS API Gateway and Lambda?
24. What is the purpose of creating a mock integration in AWS API Gateway?
25. How do you define a successful response in a mock integration?